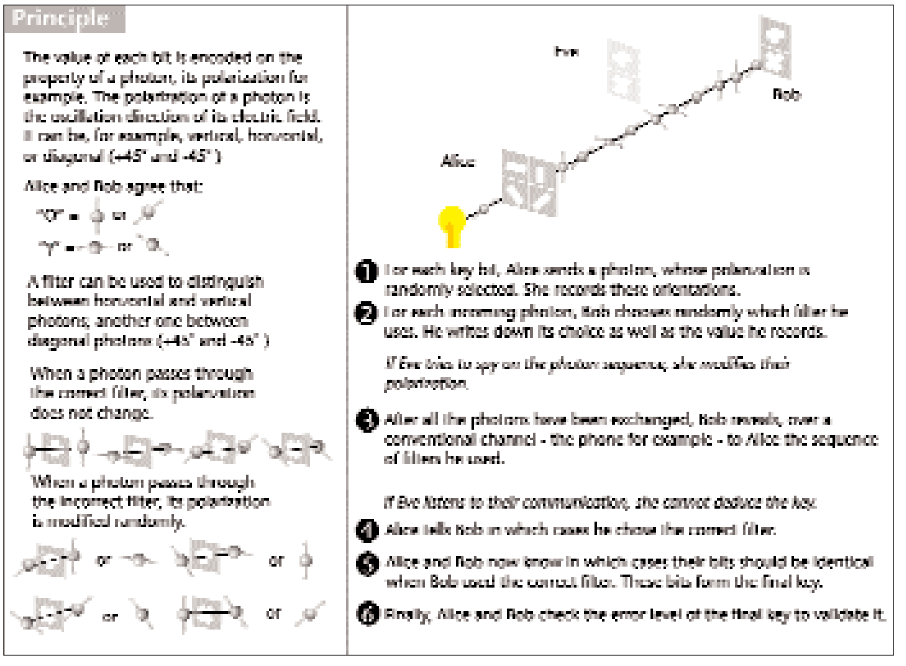

Figure 5: *Operation of Quantique's QKD technology*

Uncertainty Principle, you can be certain of either the momentum or the location of a particle, but not both. Measuring one will inevitably alter the other. This implies that if Alice sends Bob one quantum particle in a known state, and a snooper attempts to read that state, he will alter the particle in some measurable way.

The state of the art for key distribution is Quantum Key Distribution (QKD). QKD involves transmitting the bits of a key as qubits represented by quantum particles. Photons are the choice in current commercial products, since they can easily be carried over familiar media such as fibre optic cables. After all, a photon is just light. MagiQ and Quantique are pioneering companies with commercial QKD technology available now. Both claim to have deployed in some large companies and defence installations. Both are currently limited to tightly controlled fibre optic installations with range under 60 km. British Defence researchers have managed to transmit usable photons across 23 km of open air, and hope to eventually be able to bounce them off satellites hundreds of kilometres up for global coverage.

Sites to Visit

Los Alamos Quantum Cryptography
www.idquantique.com
Getting started with quantum computing
http://www.qubit.org/library/introductions.html
Interactive Young's Experiment for Quantum Superposition
http://www.colorado.edu/physics/2000/schroedinger/
[Originator of Quantum Computing Theory]
http://www.qubit.org/people/david/David.html
Belcher's Research on using Virus to make Quantum Dots
http://www.cm.utexas.edu/belcher/Research/
Introduction to Quantum Cryptography
http://www.cs.dartmouth.edu/~jford/crypto.html
[Commercial Quantum Key Distribution kits]
http://www.idquantique.com/

Applying the brakes

While there is significant progress being made in the field of quantum computing, researchers are still grappling with some fundamental problems in the fabrication of actual computers. One of the most difficult problems is that of Decoherence. For a qubit to function usefully, it must be completely isolated from its surroundings or the 'environmental noise' will cause decay of the quantum state. As yet, qubits have been maintained only for fractions of a second at a time. However, this situation is improving steadily.

The second major problem is that of fabrication. At quantum scales, light cannot be used to etch the control structures required. It is too broad a brush. Electron-beam lithography has been used to create quantum structures on conventional semiconductors in NEC, Japan earlier this year. This is a significant breakthrough, because it demonstrated the integration of quantum computing technology with existing semiconductor-based systems. Scientists at the University of New South Wales in Australia have fabricated, what appears to be the first mass producible silicon-chip qubit that can be read by sensitive transistors. They expect to have it performing useful processing but 2007.

A more far-fetched, yet apparently successful, attempt to build layers of quantum dots using a virus was made at the University of Texas. Scientists managed to create modified viruses that captured Zinc Sulphide molecules to form quantum dots. The viruses were then made to self-organise into a sheet of regular crystalline structure with uniformly spaced quantum dots. Work remains to be done to connect the dots, but the scientists are optimistic.

While these are the major roadblocks on the way to a successful commercial quantum computer, there is theoretical evidence to show that quantum computers may be energy guzzlers, consuming megawatts of electricity to keep two steps ahead of Decoherence. Better materials and techniques to reduce Decoherence would reduce energy consumption.

Throwing some light

In 1982, Richard Feynman indicated the possibility of quantum computers. That's when work began to prove it was possible, and figure out the theory behind its workings. David Deutsch, a scientist associated with Oxford University was awarded the Paul Dirac Prize for 'pioneering work in quantum computation leading to the concept of a quantum computer...' in 1998.

It took 50 years for computers to progress from gigantic building filling monsters to compact boxes on the desktop, that led the information revolution. Quantum computing is in its first decade yet, and already many basic problems considered insurmountable have been cracked. While we suggest you not line-up for a quantum *QuakeX* tournament anytime soon, it is quite a fairly safe bet that quantum computers with 'Qintel Inside' will be powering our not so distant future.

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